
Handling of large context in LLMs

Hardik Joshi

M.Tech AI

22886

hardikjoshi@iisc.ac.in

Abstract

The LLM context-length problem emanates from its inability to cope with long-range dependency appropriately because of the quadratic complexities attention mechanisms incur in traditional transformer designs. The problem rears its head when attention mechanisms are needed to pass through long documents or longer conversations. Solutions like Transformer-XL, which introduces recurrence for longer memory retention and the Compressive Transformer compressing past memory for much more efficient storage, look at ways to improve in this respect. Another application of Hierarchical Attention Mechanism and Hierarchical Attention Transformer Networks is through use of hierarchical structures to capture interdependencies at different levels of abstraction. This paper compares such models to evaluate their performances in overcoming the context length problem and determine which is more suitable for a specific NL processing task involving long sequences.

1 Introduction

The rapid progress in the large language models (LLMs) has significantly improved them in a wide range of NLP tasks, starting from text generation to the machine translation and document classification. However, one most fundamental challenge that remains till now is the context length problem, which refers to the model's inability to have an efficient processing of long-range dependencies due to the limited sequence length and quadratic scaling attention mechanisms. This limitation becomes particularly evident in traditional transformers in that the attention mechanism in the model becomes computationally expensive with long input sequences. As such, coherence and context often become an issue for models over extended texts, such as lengthy documents or conversations.

Several novel architectures have been proposed to counter this problem. One of the pioneering solutions is Transformer-XL, which introduces recurrence in the attention mechanism so that the model can maintain memory across different segments of text and therefore can extend the context length without an increase in computational cost. This approach makes it possible for the model to handle longer sequences by capturing long-term dependencies between distant tokens. It does this more than any of the former transformers where, along with keeping the history memory, the memory compresses itself to avoid taking away crucial information yet consuming a large amount of resources. More hierarchical techniques include HAM and HATN to deal with the length problem of contexts, using hierarchy in designing the sequence for input. These models operate on input at different abstraction levels and allow them to concentrate on important parts of text while keeping the long range of dependencies within the whole sequence. Aggregating information up higher levels of hierarchy is a means for these models to scale effectively to larger inputs without a loss of performance.

The paper attempts a detailed comparison of Transformer-XL, Compressive Transformer, HAM, and HATN. Specifically, we evaluate Transformer-XL and Compressive Transformer on a text generation task, analyzing their ability to generate coherent and contextually accurate sequences over long

Table 1: Results on Wiki9 Dataset

Model	Attention Heads	Perplexity (PPL)
Transformer-XL	2	35.1142
Transformer-XL	8	34.0151
Compressive Transformer	2	34.3316
Compressive Transformer	8	32.4866

Table 2: Results on BookCorpus Dataset

Model	Attention Heads	Perplexity (PPL)
Transformer-XL	2	35.1142
Transformer-XL	8	34.0151
Compressive Transformer	2	34.3316
Compressive Transformer	8	32.4866

texts. Additionally, we assess the performance of all four models on document classification tasks, where long-context understanding is essential for accurate categorization. Thus, by comparison, we conclude the merits and flaws of each technique towards its applications in practice for the practical handling of long-context NLP tasks.

2 Problem Statement

The context length problem in large language models arises from the inability of these models to capture long-range dependencies in input sequences. Traditional attention mechanisms limit their ability to do so, and as the length of input sequences increases, standard Transformer models experience a severe degradation in performance in terms of memory usage and computational complexity, making them inefficient for tasks requiring long-context processing. This challenge manifests particularly in text generation and document classification tasks, for which coherence over long sequences or accurate classification of lengthy documents is critical.

Existing models such as Transformer-XL, Compressive Transformer, Hierarchical Attention Mechanism (HAM), and Hierarchical Attention Transformer Networks (HATN) provide alternative means to address the context length problem, but their comparative effectiveness in different tasks has not been well explored. The problem, therefore, is to evaluate and compare the ability of these models to address the context length issue, specifically in the context of text generation and document classification tasks. The goal is to assess their performance in maintaining long-range dependencies, model efficiency, and task accuracy to determine which model is most suitable for handling long-context challenges in NLP.

3 Result

3.1 Comparison of Transformer-XL and Compressive Transformer

The performance of Transformer-XL and Compressive Transformer performance was tested on the BookCorpus and Wiki9 datasets, under two different configurations for the number of attention heads, namely 2 and 8. The evaluation metric used is the perplexity, PPL, where the lower value is better. The experiments were carried out under uniform conditions and common configurations included a batch size of 16, 100 training batches, 4 memory units, a compression rate of 3, and the Adam optimizer with a learning rate of 0.0001.

3.2 Comparison of Transformer-XL, Compressive Transformer, Hierarchical Attention Mechanism, and HATN

We examined the performance of Transformer-XL, Compressive Transformer, Hierarchical Attention Mechanism (HAM), and Hierarchical Attention Transformer Networks (HATN) over the newsgroups

Table 3: Results on newsgroups Dataset

Model	Accuracy (%)
Transformer-XL	38.78
Compressive Transformer	41.47
Hierarchical Attention Mechanism GRU (HAM)	42.09
Hierarchical Attention Transformer Networks (HATN)	44.23

available on the Kaggle database for the document classification task, where accuracy is the primary metric. The experiments were carried out under uniform conditions and common configurations included a batch size of 1, 90 training batches for each class, 4 memory units, a compression rate of 3, and the Adam optimizer with a learning rate of 0.0001. **Note:-** The code of HANT and HAM are official code provided by author.

4 Result Analysis and Discussion

4.1 Comparison of Transformer-XL and Compressive Transformer

The results on the Wiki9 and BookCorpus datasets show that the Compressive Transformer performs better than the Transformer-XL model in terms of perplexity (PPL), which is the evaluation metric used.

For the Wiki9 dataset, the Compressive Transformer with 8 attention heads achieves the lowest PPL of 32.4866, outperforming the Transformer-XL with 8 attention heads (PPL of 34.0151). A similar trend is observed on the BookCorpus dataset, with the Compressive Transformer with 8 attention heads achieving the lowest PPL of 32.4866 compared to the Transformer-XL’s best performance of 34.0151.

The lower perplexity scores indicate that the Compressive Transformer is better at modeling the long-range dependencies in the text, leading to more coherent and contextually accurate sequence generation compared to the Transformer-XL. The key advantage of the Compressive Transformer seems to be its ability to effectively compress past memory, allowing it to maintain long-term context while being more memory and computationally efficient than the recurrence-based approach of Transformer-XL.

4.2 Comparison of Transformer-XL, Compressive Transformer, Hierarchical Attention Mechanism (HAM), and Hierarchical Attention Transformer Networks (HATN)

On the newsgroups document classification dataset, the Hierarchical Attention Transformer Networks (HATN) model achieves the highest accuracy of 44.23%, outperforming the other three models. The Hierarchical Attention Mechanism (HAM) model comes in second with an accuracy of 42.09%, followed by the Compressive Transformer at 41.47% and the Transformer-XL at 38.78%.

The superior performance of the HATN model suggests that the hierarchical attention mechanism, which operates on the input at different levels of abstraction, is more effective at capturing the long-range dependencies and important parts of the text, leading to better document classification accuracy. The Compressive Transformer and Transformer-XL, while effective at long-sequence modeling, may still struggle to fully capture the complex interdependencies present in lengthy documents required for accurate classification.

The results indicate that the hierarchical attention-based models like HATN are more suitable for tasks that require a deep understanding of long-context information, such as document classification, compared to the recurrence-based (Transformer-XL) or memory compression-based (Compressive Transformer) approaches.

4.3 Conclusion

Overall, the results suggest that the Compressive Transformer and Hierarchical Attention Transformer Networks (HATN) are the most promising solutions for addressing the context length problem in

111 LLMs, outperforming the Transformer-XL and Hierarchical Attention Mechanism (HAM) models
112 across the evaluated tasks. The Compressive Transformer’s memory compression technique and the
113 HATN’s hierarchical attention mechanism appear to be effective strategies for handling long-range
114 dependencies in text. The choice between the two models would depend on the specific requirements
115 of the NLP task at hand.

116 5 Reference

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